STUDENT RESULT ANALYZER

A MINI PROJECT REPORT

Submitted by

AJMAL V B (LPKD22CS068)

ARYA V P (PKD22CS021)

DIYA ROSE JOJI (PKD22CS030)

FARSANA K K (LPKD22CS070)

to

the APJ Abdul Kalam Technological University

Department of Computer Science and Engineering

Government Engineering College Sreekrishnapuram, Palakkad

APRIL, 2025

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VISION AND MISSION OF THE COLLEGE

VISION

Excellence through the wings of science and technology.

MISSION

To transform youth to talented engineers with creativity and integrity who can meet the technological challenges for the service of society.

VISION AND MISSION OF THE DEPARTMENT

VISION

To become a Centre of Excellence in Computing and allied disciplines.

MISSION

To impart high quality education in Computer science and Engineering that prepares the students for

rewarding, enthusiastic and enjoyable careers in the industry, academia and other organizations.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Apply good analytic, design, and implementation skills required to formulate and solve computer science problems.

PEO2: Be employed as computer science professionals beyond entry-level positions.

PEO3: Be able to route their talents in to post graduate and research programs, promoting remarkable advancements in emerging areas.

PEO4: Demonstrate that they can function, communicate, collaborate and continue to learn effectively as ethically and socially responsible computer science professionals.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Problem Solving Skills**:** Ability to design and develop computer programs and computer based systems of moderate complexity in the areas pertaining to system software, multimedia, database, networking, artificial intelligence, web- design and information security.

PSO2: Professional Skills: Ability to apply standard practices and methods in software project management and software development using suitable programming environments to deliver quality product for the industry.

DECLARATION

We hereby declare that the project report entitled ”**STUDENT RESULT ANALYZER**” submitted by us to the APJ Abdul Kalam Technological University during the academic year 2024- 25 in partial fulfilment of the requirements for the award of Degree of Bachelor of Technology in COMPUTER SCIENCE AND ENGINEERING is a record of bonafide project work carried out by us under the guidance and supervision of Assistant Prof., ATHIRA P K . We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other University.

AJMAL V B (LPKD22CS068)

ARYA V P (PKD22CS021)

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FARSANA K K (LPKD22CS070)

Place: Sreekrishnapuram
Date: April 4, 2025

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
GOVERNMENT ENGINEERING COLLEGE PALAKKAD

Sreekrishnapuram, Palakkad-678633

CERTIFICATE

This is to certify that the report entitled "**STUDENT RESULT ANALYZER**"submitted by **AJMAL V B** (LPKD22CS068)**, ARYA V P** (PKD22CS021), **DIYA ROSE JOJI** (PKD22CS030)**, FARSANA K K** (LPKD22CS070) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Guide
ATHIRA P K
Assistant Professor, Dept. of CSE

Head of the Department
Dr. SABITHA S
Professor, Dept. of CSE

Project Coordinators
KALA M T
MUMTHAS OY

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Thank you all for being part of this journey.

Team Student Result Analyzer

ABSTRACT

The *Student Result Analyzer* is a web-based application designed to streamline the process of academic result management and performance analysis. Developed using HTML, CSS, JavaScript, PHP, and MySQL, this system enables secure storage, retrieval, and visualization of student results. It offers role-based access, allowing teachers to add or update marks and generate performance insights using interactive charts, while parents and students can track academic progress through user-friendly visual reports. Key features include PDF-to-Excel result conversion, graphical analysis via Chart.js, and a responsive UI built with Bootstrap. The project aims to reduce manual workload, improve data accuracy, and enhance transparency in result management. Future enhancements include integrating AI-based performance prediction, an automated grading system, and mobile application support to further expand accessibility and functionality.

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CHAPTER 1
INTRODUCTION

In today's digital era, managing and analyzing academic performance has become a crucial aspect of educational institutions. Traditional methods of result processing—often manual and paper-based—are time-consuming, error-prone, and lack real-time insights. Recognizing these challenges, the *Student Result Analyzer* was developed as a comprehensive web-based platform aimed at transforming the way academic data is handled.

This system offers a centralized and automated approach to storing, managing, and visualizing student performance data. It is built using modern web technologies such as HTML, CSS, JavaScript, PHP, and MySQL, ensuring a robust and interactive user experience. The core objective is to simplify the tasks of teachers in entering and analyzing marks, and to empower parents and students with instant access to results and performance trends.

Through secure logins and role-based access, the platform accommodates multiple stakeholders—teachers, students, and parents—while maintaining data privacy and integrity. Teachers can upload and update student marks effortlessly, and generate dynamic graphs that provide clear insights into individual and overall academic progress. Parents and students can view the same data in an intuitive format, helping them stay informed and engaged with academic development.

An additional feature includes the ability to convert result data from PDF to Excel format, facilitating easier data manipulation and upload. By leveraging visual analytics with tools like Chart.js, the platform presents performance data in an easily understandable manner, highlighting patterns, improvements, and areas that need attention.

The *Student Result Analyzer* not only reduces administrative burden but also enhances decision-making and academic planning through timely and transparent result analysis. It marks a significant step towards digital transformation in the education sector, paving the way for future advancements like AI-powered performance prediction and mobile accessibility.

1.1 SCOPE

The *Student Result Analyzer* is designed to serve as an all-in-one academic performance monitoring and result management system tailored for educational institutions. Its scope encompasses a wide range of functionalities, stakeholders, and potential use cases, making it an essential tool for modern academic administration.

At its core, the system allows teachers to securely manage student results by providing functionalities to add, edit, and update marks with ease. The system replaces conventional manual processes with a digital interface, significantly reducing time consumption and minimizing errors in result management. Teachers can also generate and analyze graphical representations of student performance, allowing for data-driven decision-making and better student mentoring.

For students and parents, the system offers real-time access to academic results through an intuitive, role-based portal. This increases transparency and encourages continuous engagement with academic performance. The visual graphs make it easier for non-technical users to understand trends, strengths, and areas for improvement.

The platform's architecture ensures data integrity and security by using secure login systems and backend validations. It supports uploading results through both manual input and PDF to Excel conversion, which streamlines data entry, especially when dealing with bulk results.

The *Student Result Analyzer* also provides performance visualization using Chart.js, enabling detailed insights into semester-wise and subject-wise progress. This helps in identifying academic trends at both individual and class levels, supporting teachers and institutions in tailoring their instructional strategies accordingly.

From a technical perspective, the system utilizes HTML, CSS, and JavaScript for the front end, PHP for backend logic, and MySQL for secure database management. With the use of responsive design principles and Bootstrap, the application ensures a smooth experience across different devices and screen sizes.

Furthermore, the scope extends beyond its current functionality with planned future enhancements, including:

- AI-driven performance prediction,
- Automated grading systems,
- Integration of a dedicated mobile application to increase accessibility for all users.

In summary, the *Student Result Analyzer* not only addresses current challenges in academic result management but also lays a strong foundation for scalable and intelligent educational analytics systems in the future.

CHAPTER 2
LITERATURE SURVEY

With the growing emphasis on data-driven education, the need for automated student result analysis systems has gained significant momentum. Traditional methods of analyzing academic performance often involved manual tabulation of marks and subjective interpretation, which were time-consuming and prone to human error. Over the years, educational institutions have begun shifting toward digital platforms that offer streamlined, efficient, and transparent performance evaluation. Early systems focused on basic mark management, while more recent developments have introduced result analysis dashboards that present insights through graphs, charts, and comparative metrics. These systems not only help educators monitor student progress but also assist in identifying areas where learners struggle, enabling timely academic interventions.

Research in this domain has emphasized the integration of technologies such as database systems for secure and structured data storage, along with frontend frameworks for interactive visualization. Several studies have also highlighted the importance of student-centric design that ensures clarity and ease of use for both faculty and students. Platforms have been developed that track semester-wise performance, calculate pass percentages, and evaluate subject-wise trends to enhance institutional decision-making. The evolution of these systems reflects the growing awareness of leveraging academic data to foster educational excellence. The insights gained from previous research have significantly influenced the design and development of the **Student Result Analyzer**, which aims to provide a comprehensive, intuitive, and reliable platform for academic result evaluation.

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2.1: Automated Student Result System

Authors: John & Smith (2020)

This paper presents the development of a basic Excel-based student result storage and processing system aimed at simplifying academic grade management. The system automates the calculation of grades based on predefined rules and grading criteria, offering institutions a digital alternative to manual mark sheets. While it effectively reduces manual workload and ensures consistency in result calculations, the system lacks critical features such as analytical comparisons, data filtering, and broader insights into student performance. Its reliance on Excel limits scalability and restricts data handling when dealing with large student populations or multiple academic batches.

How Our Project is Better:

The Student Result Analyzer advances this concept by using a MySQL-based structured database, which provides efficient data storage and management for larger academic datasets. The system supports subject-wise, batch-wise, and individual student result comparisons, enabling detailed performance analysis across different levels. Additionally, the use of a web-based interface enhances user accessibility and allows for real-time interaction with result data, making it more effective for institutional review and academic planning.

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2.2: Digital Report Cards Using AI

Authors: Sharma et al. (2019)

This paper explores the development of an AI-based system designed to generate personalized student performance reports. By analyzing historical academic data, the system leverages predictive algorithms to forecast future grades, offering students tailored feedback on their academic progress. The individualized nature of the reports allows for focused improvement strategies. However, the system lacks broader analytical capabilities, such as evaluating overall performance trends across batches, departments, or academic years. This limitation reduces its effectiveness in supporting administrative decision-making and academic planning on a larger scale.

How Our Project is Better:

The Student Result Analyzer shifts the focus from individual predictions to institution-wide performance analysis. It enables users to filter and analyze results department-wise, batch-wise, and subject-wise, offering a broader understanding of academic trends. The system includes visual representations of data that help identify high-performing and underperforming areas, supporting strategic improvements in teaching and curriculum delivery. This holistic approach ensures that the tool is valuable not only for students but also for faculty and administrators.

2.3: Web-Based Student Performance Tracker

Authors: Lee & Kim (2021)

This paper introduces a web-based application developed to display student performance data in a structured tabular format. The system incorporates static graphs to provide a visual overview of student results, aiming to enhance the readability of academic reports. While functional for basic performance tracking, the system lacks interactivity—users cannot apply filters to view specific subsets of data such as by semester or subject. Additionally, the absence of CSV upload functionality makes bulk data entry time-consuming and inefficient, especially for institutions managing large student populations.

How Our Project is Better:

The Student Result Analyzer addresses these limitations by enabling bulk result uploads through CSV files, significantly streamlining the data entry process. It also supports interactive filtering, allowing users to sort and analyze results by subject, semester, department, or batch. The system dynamically generates performance reports, offering real-time insights that are both user-friendly and adaptable to institutional needs. This flexibility enhances the overall usability and efficiency of academic result management.

2.4: Online Result Portal for Universities

Authors: Fernandez et al. (2020)

This paper presents the development of an online portal designed to generate semester-wise academic reports for university students. The system simplifies the process of publishing and accessing individual semester results, providing a convenient platform for both students and administrators. However, the portal is limited in its analytical scope—it does not support tracking a student's academic growth over multiple semesters, nor does it provide tools for long-term performance evaluation or trend analysis across departments or batches.

How Our Project is Better:

The Student Result Analyzer expands on this concept by offering longitudinal tracking of student performance across semesters, enabling detailed progress evaluation. The system allows users to compare results within a batch or department, helping identify collective academic strengths and weaknesses. It also incorporates graphical visualizations that present long-term academic trends in an intuitive format, making it easier for educators to monitor overall growth and implement data-driven improvements.

2.5: Smart School Management System

Authors: Bose & Kumar (2018)

This paper discusses the implementation of a school management platform focused on simplifying administrative tasks such as grade entry and report card generation. While effective in automating routine operations, the system is primarily limited to basic functionalities like recording marks and determining pass/fail status. It lacks interactive analytics capabilities and does not allow users to filter or view data based on specific academic criteria such as subject, batch, or performance trends.

How Our Project is Better:

The Student Result Analyzer enhances functionality by offering detailed performance insights across subjects, batches, and semesters. It supports the generation of customized reports, which can be downloaded in multiple formats like PDF and Excel for administrative use. The system also features real-time filtering, enabling users to generate tailored views and analyze academic data from various perspectives, making it a more powerful and adaptable tool for modern educational institutions.

2.6: University Exam Result Automation

Authors: Ravi & Das (2020)

This paper outlines the development of an automated system aimed at generating individual student result reports for university examinations. The system streamlines the result publication process by organizing student marks in a tabular format, ensuring clarity and ease of access. However, it lacks features for comparative analysis across batches, subjects, or departments. The absence of visual or trend-based analytics also limits its usefulness for institutional performance review and strategic planning.

How Our Project is Better:

The Student Result Analyzer goes beyond individual reporting by enabling batch-wise and department-wide performance analysis. It integrates graphical data visualizations, which offer clearer and more intuitive insights into student outcomes. Furthermore, the system supports trend analysis across multiple semesters, helping stakeholders monitor academic progress and implement data-driven improvements effectively.

PAPER NAME	AUTHOR(S)	ALGORITHM	ADVANTAGE	DISADVANTAGE
Automated Student Result System	John & Smith (2020)	Excel-based Calculation	Simplifies grade calculations; digital result storage	No visualization; limited comparative analysis
Digital Report Cards Using AI	Sharma et al. (2019)	Predictive AI Models	Personalized reports with grade predictions	Lacks batch-level trend analysis
Web-Based Student Performance Tracker	Lee & Kim (2021)	Static Web Framework	Online result display with basic graphs	No interactive filtering; no bulk upload support
Online Result Portal for Universities	Fernandez et al. (2020)	Web App with DB Support	Generates semester-wise reports	No multi-semester tracking or progress monitoring
Smart School Management System	Bose & Kumar (2018)	Basic Rule-based System	Automates grade entry and report cards	Only pass/fail analytics; no filtering by criteria
University Exam Result Automation	Ravi & Das (2020)	Automated Report Engine	Quick generation of individual reports	No comparative tracking; lacks graphical or trend analysis

Table 2.1: Literature Survey Summary

CHAPTER 3
PROBLEM STATEMENT

In many educational institutions, the process of managing and analyzing student academic results remains largely manual or minimally digitized. Teachers often rely on spreadsheets or paper-based methods to record marks, calculate performance metrics, and generate progress reports. These methods are time-consuming, prone to human error, and lack scalability. Furthermore, students and parents face difficulty in accessing results promptly and understanding performance trends due to the absence of clear and interactive visualizations.

Additionally, traditional result management systems often do not offer features like performance analytics, secure access control, or historical comparison of academic progress. This lack of data-driven insight limits the ability of educators to identify struggling students early, provide targeted support, and ensure academic excellence.

There is a clear need for a centralized, user-friendly platform that can automate result management, provide secure and role-based access, and present academic performance in a visually intuitive manner. A system that supports bulk result uploads, real-time analysis, and easy accessibility for all stakeholders can significantly enhance the academic monitoring process.

The *Student Result Analyzer* project addresses this gap by offering a web-based solution that simplifies result entry, ensures data accuracy, and delivers insightful performance visualizations—empowering teachers, students, and parents to make informed academic decisions.

3.1 PROBLEM STATEMENT

Manual result management in schools and colleges is time-consuming, error-prone, and lacks real-time performance insights. Students and parents often struggle to access and understand academic progress. There is a need for a digital solution that automates result processing, ensures secure access, and provides clear visual performance analysis.

3.2 OBJECTIVES

1. **Automate Result Management** – Eliminate manual data entry by enabling easy addition and updating of student marks through a web interface.
2. **Provide Secure Access** – Implement role-based login for teachers, students, and parents to ensure data privacy and control.
3. **Generate Progress Reports** – Offer detailed and visually engaging performance reports for individual students and entire classes.
4. **Enable PDF to Excel Conversion** – Simplify bulk data uploads by converting result data from PDF format to structured Excel files.
5. **Visualize Academic Performance** – Use interactive charts and graphs to help stakeholders easily understand academic trends and outcomes.

CHAPTER 4

DESIGN

This chapter presents the structural, visual, and functional design of the **Student Result Analyzer** system—an intelligent platform developed to streamline the evaluation, analysis, and presentation of student academic performance. The system is intended to support educational institutions in transitioning from manual result processing to a more automated, accurate, and visually driven analytical framework.

The design is focused on improving both usability for academic staff and transparency in result reporting. It incorporates key software engineering principles such as modularity, scalability, maintainability, and data integrity, ensuring that the system can adapt to various data formats, class sizes, and institutional policies.

Key features include:

- Automated grade calculations from uploaded student result sheets.
- Visual representation of subject-wise and student-wise performance.
- Generation of downloadable reports for academic reviews.
- Optional integration of machine learning models for trend prediction and performance insights.

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4.1 SYSTEM ARCHITECTURE

The **Student Result Analyzer** is built using a modular and layered architecture that ensures data consistency, efficient processing, and a smooth user experience. The system includes three primary modules:

1. Input Module

- **User Authentication:**
  Secure login for admin or faculty to access features like data upload and result analysis.
- **Data Entry / File Upload:**
  Users can upload CSV or Excel sheets containing student marks, names, subjects, and other metadata. These are parsed for analysis.

2. Processing Module

- **Data Processing Engine:**
  Validates and normalizes student data, computes grades, pass/fail ratios, average marks, and class performance trends.
- **Analytics Module:**
  Performs deeper analysis such as subject-wise performance, topper identification, low-performance detection, and graphical result comparisons.

3. Output Module

- **Visualization Dashboard:**
  Displays results using tables, pie charts, and bar graphs for better insights into performance metrics.
- **Report Generation:**
  Auto-generates downloadable PDF/Excel reports for each class, subject, or student.

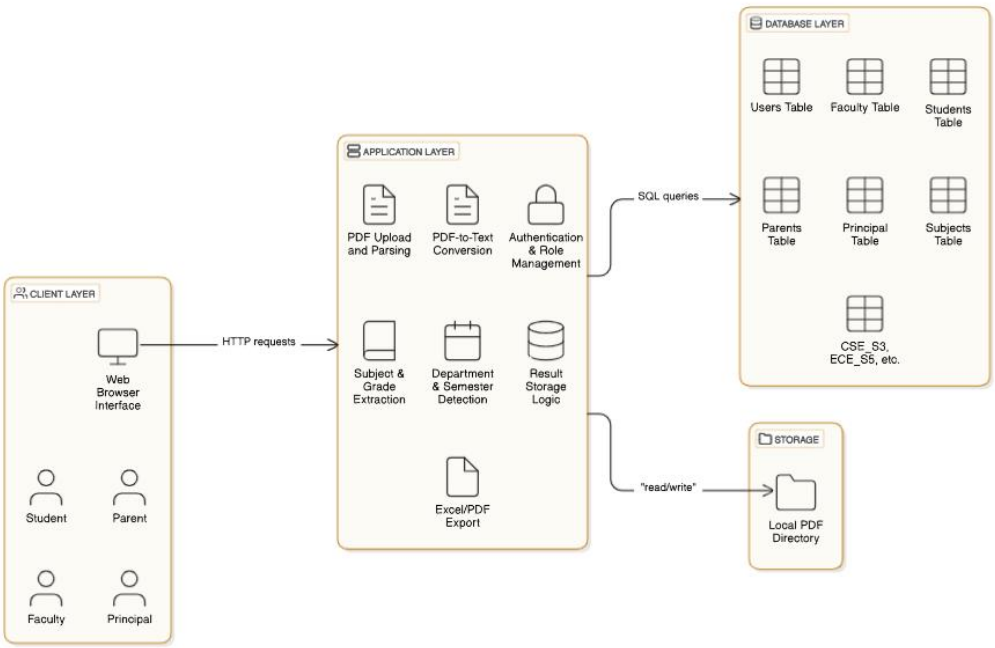


Figure 4.1: Architecture Diagram

4.2 USE CASE DIAGRAM

Actors:

- **Teacher**
- **Student**
- **Parent**
- **System Administrator (optional)**

Use Cases:

1. **Login**
    - Actors: Teacher, Student, Parent
    - Description: Secure login with role-based access
2. **Add/Update Results**
    - Actor: Teacher
    - Description: Enter or modify student marks
3. **View Results**
    - Actors: Student, Parent
    - Description: View detailed student performance
4. **Generate Performance Graphs**
    - Actors: Teacher, Student, Parent
    - Description: Visual representation of marks using charts
5. **Download Reports**
    - Actors: Teacher, Student, Parent
    - Description: Download performance reports in PDF/Excel format
6. **PDF to Excel Conversion**
    - Actor: Teacher
    - Description: Convert result data from PDF to Excel for easy upload
7. **Manage User Accounts**
    - Actor: System Administrator (if implemented)
    - Description: Add or remove users and manage access permissions

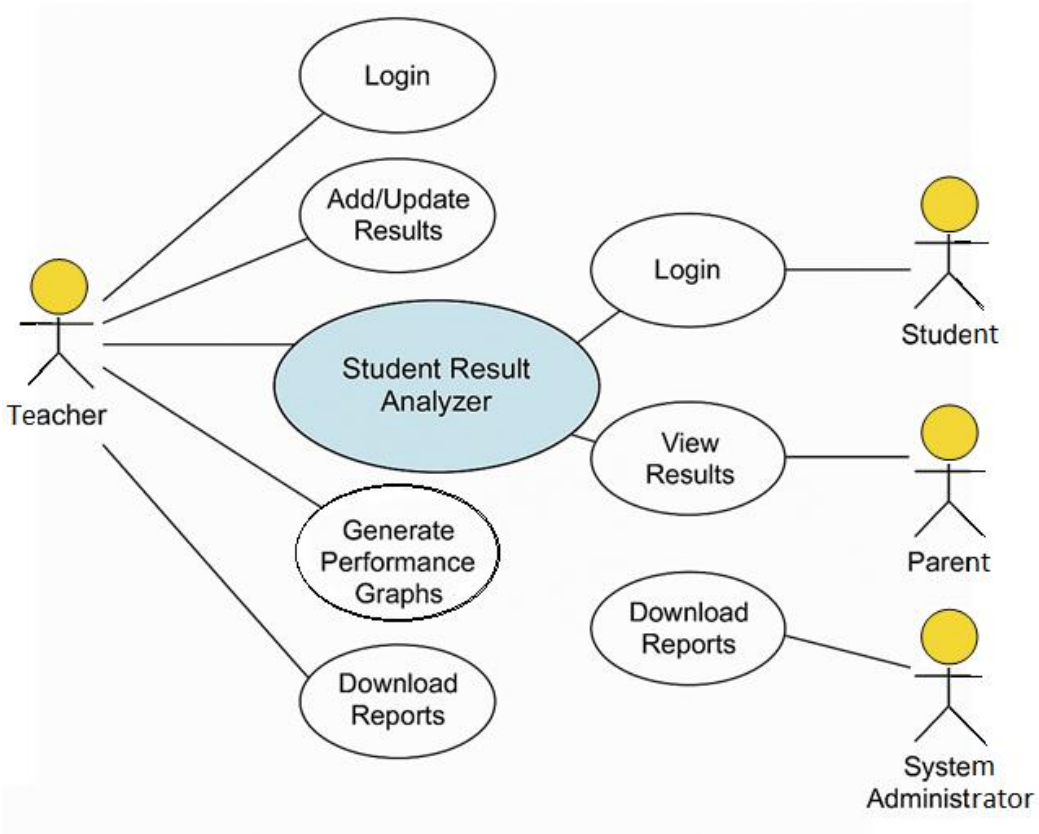


Figure 4.2: Use Case Diagram

4.3 SYSTEM REQUIREMENTS

To ensure optimal performance and seamless integration of all modules in the Student Result Management system, the following software requirements have been identified:

1. Frontend Requirements

The frontend is responsible for the user interface and interaction layer of the system. It is developed using standard web technologies and tools.

- **Languages & Technologies**:
  - HTML5 – for structuring web pages
  - CSS3 – for styling and layout
  - JavaScript – for interactive elements and dynamic behaviours
- **Frameworks & Libraries**:
  - Bootstrap – for responsive design and pre-built UI components
  - Chart.js – for rendering interactive graphs and performance charts
- **Browser Support**:
  - Google Chrome (latest version)
  - Mozilla Firefox
  - Microsoft Edge
  - Safari
- **Screen Resolution**:
  - Optimized for 1366×768 and higher

2. Backend Requirements

The backend handles the core logic, database management, and server-side scripting for data processing.

- **Languages & Technologies**:
  - PHP 7.4 or higher – for server-side scripting and business logic
- **Database Management System**:
  - MySQL 5.7 or higher – for structured data storage and queries
- **Server Requirements**:
  - XAMPP/WAMP/LAMP Stack
  - Apache HTTP Server – for hosting the application
- **Operating System (Recommended for Local Hosting)**:
  - Windows 10/11 or Ubuntu 20.04+
  - Minimum RAM: 4 GB
  - Processor: Intel i3 or higher

CHAPTER 5

WORKPLAN

This chapter outlines the structured methodology adopted for the development of the **Student Result Analyzer** system. It details the various phases of implementation, systematic task distribution among team members, and a timeline for efficient project execution. The goal was to ensure smooth collaboration, timely delivery, and quality assurance throughout the development lifecycle

5.1 **GANTT CHART**

The Gantt Chart serves as a visual representation of the entire project schedule. It breaks down the complete lifecycle into major tasks and sub-tasks, assigns estimated time frames for each phase, and maps dependencies where applicable. The chart provides a clear roadmap from initial research to final deployment and documentation..

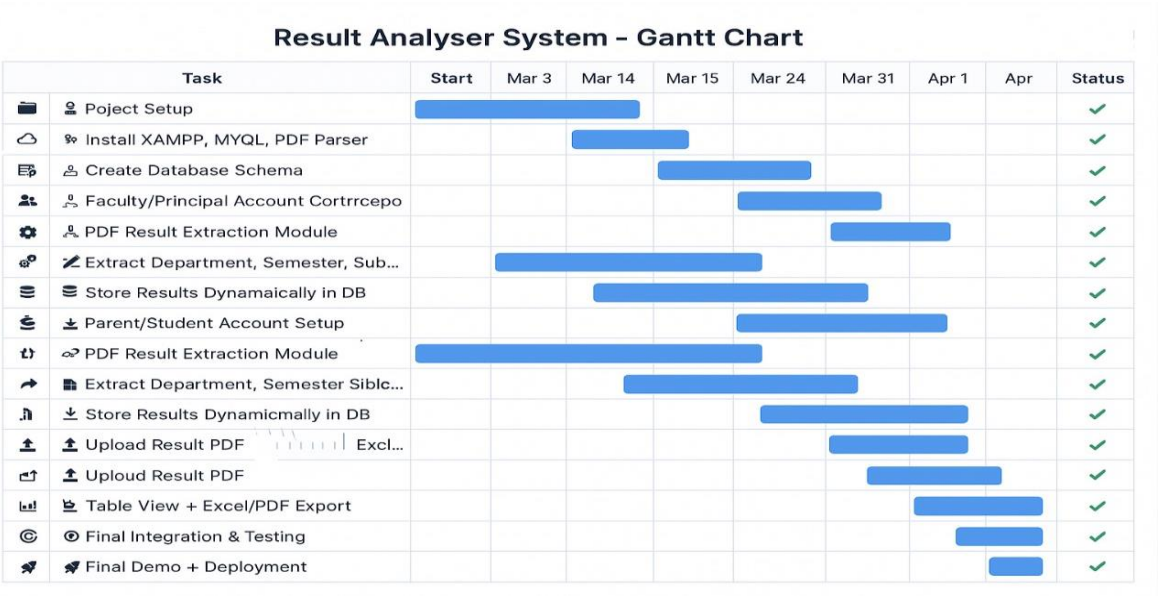


Figure 5.1: Ganttchart

5.2 TASK ALLOCATION

Efficient task allocation is crucial to streamline workflow and ensure that each team member contributes according to their expertise. The responsibilities were delegated as follows:

TASK	Ajmal V B	Arya V P	Diya Rose Joji	Farsana K K
Literature Survey	Automated Student Result System	Digital Report Cards Using AI	Web-Based Student Performance Tracker	Online Result Portal for Universities
Formulation & Objectives	Done	Done	Done	Done
Design	UI Design	Architecture Design	Use case diagram	Workflow Diagram
Preliminary Analysis	Done	Done	Done	Done
Phase 1 Report	Done	Done	Done	Done
Implementation	Frontend & UI/UX	Backend & Database Integration	Data Processing Logic	Testing & Deployment
Phase 2 Report	Done	Done	Done	Done

Table 5.2: TaskAllocationTable

CHAPTER 6

IMPLEMENTATION

This chapter elaborates on the actual implementation process of the Student Result Management system, focusing on frontend and backend development , database and authentication systems, as well as deployment procedures. Each subsection outlines the tools, frameworks, and methods used to bring the platform Student Result Management system to life.

6.1 SOFTWARE IMPLEMENTATION

6.1.1 FRONTEND DEVELOPMENT

The frontend is responsible for the user interface and interaction. It is developed using standard web technologies that ensure responsiveness, accessibility, and a user-friendly experience.

Technologies Used

- **HTML5** – For structuring content and layout.
- **CSS3** – For styling the UI components (colours, fonts, layout).
- **JavaScript** – For dynamic behaviour and client-side validation.
- **Bootstrap** – To ensure responsive design and grid layout.
- **Chart.js** – For drawing graphical visualizations of student performance.

Key Features Implemented

- **Role Selection Page**: Users choose between Faculty, Student, Parent, and Principal.
- **Login Forms**: Role-based login using register number or credentials.
- **Student Dashboard**: Displays subject-wise performance in both table and graph format.
- **Faculty Query Page**: Faculty can type queries in natural language and view the results in chart form.
- **Parent Page**: Shows student's profile (photo, name, register number) and results.
- **Responsive Design**: Works across desktops, tablets, and mobile devices.

Design Focus

- Simple and clean UI with pastel green colour schemes.
- Icons and layout optimized for intuitive navigation.
- Google Charts/Chart.js integrated for academic performance visualization.

6.1.2 BACKEND DEVELOPMENT

The backend handles the application logic, database interaction, data processing, and server-side render

Technologies Used

- **PHP** – For server-side scripting and business logic.
- **MySQL** – For managing the student database.
- **XAMPP/WAMP** – As a local development environment with Apache server and MySQL.

Database Design

Tables:

- `students`: Stores student profile info (name, register number, photo).
- `results`: Stores subject-wise grades for each semester and department (e.g., ce_s1, cse_s2 , etc.).
- `subjects`: Stores subject details (subject code, name, etc.).
- `grading_system`: Stores grade-to-point mappings used for SGPA/CGPA calculations.
- `users`: Stores login credentials for different roles (faculty, principal, parents).

Proper use of **foreign keys** to link student results with student profiles.

Backend Features

- **Authentication**: Validates register number or credentials depending on role.
- **Query Processor**: Interprets natural language queries (like "Topper of CSE") and translates them to SQL.
- **Data Retrieval**: Executes SELECT queries based on user input and sends data back to the frontend.
- **Data Visualization Prep**: Encodes data in JSON format to feed into Chart.js/Google Charts.
- **Input Validation & Error Handling**: Ensures clean inputs and shows friendly messages on failures.

Integration Workflow

1. **User logs in** (e.g., as a parent or faculty).
2. **PHP backend authenticates** the input from the login form.
3. Upon success, **relevant student data is fetched** from MySQL.
4. The data is returned to the frontend and **displayed using HTML/CSS**.
5. **JavaScript/Chart.js generates graphs** dynamically from the retrieved data.

6.2 DEPLOYMENT

6.2.1 MODEL DEPLOYMENT

The *Student Result Analyzer* is deployed locally using **XAMPP**, where the PHP files are hosted on the **Apache server** and data is managed through **MySQL**. The system is accessed via `http://localhost` on a browser. It can be later deployed online using platforms like **Heroku** or **AWS** for broader accessibility.

6.2.2 WEB DEPLOYMENT

Web deployment involves hosting the *Student Result Analyzer* on a local server using **XAMPP**, which includes **Apache** for running PHP scripts and **MySQL** for database management. Users can access the system via a browser using `http://localhost/`. For online access, it can be deployed on platforms like **Heroku** or **InfinityFree** with minor configuration updates

6.2.3 TESTING AND MONITERING.

The system was tested for functionality, user roles, and input validation. Manual monitoring ensured accurate data display and smooth UI. Future deployment can use tools like server logs for tracking

CHAPTER 7
RESULTS AND DISCUSSION

This chapter evaluates the performance of the **Student Result Analyzer** system by analyzing outcomes, accomplishments, and the challenges encountered during its development and implementation. The system was developed to assist academic institutions in visualizing and analyzing student performance data effectively through an intuitive, user-friendly web interface. It emphasizes how each module contributed toward building a reliable and responsive result analysis platform.

7.1 RESULTS

The **Student Result Analyzer** successfully met its objective of providing an accessible and efficient platform for analyzing academic results and generating insightful visual reports. The system effectively executed all core functionalities, and the following key outcomes were observed:

- **Data Upload Interface:** Faculty members and administrators could easily upload student results in bulk through Excel or CSV formats.
- **Result Categorization & Filtering:** The system enabled filtering results by semester, subject, course, and department to streamline data exploration.
- **Visualization Module:** Graphs and charts (e.g., bar charts, pie charts) visually represented pass/fail percentages, subject-wise performance, and year-wise trends.
-

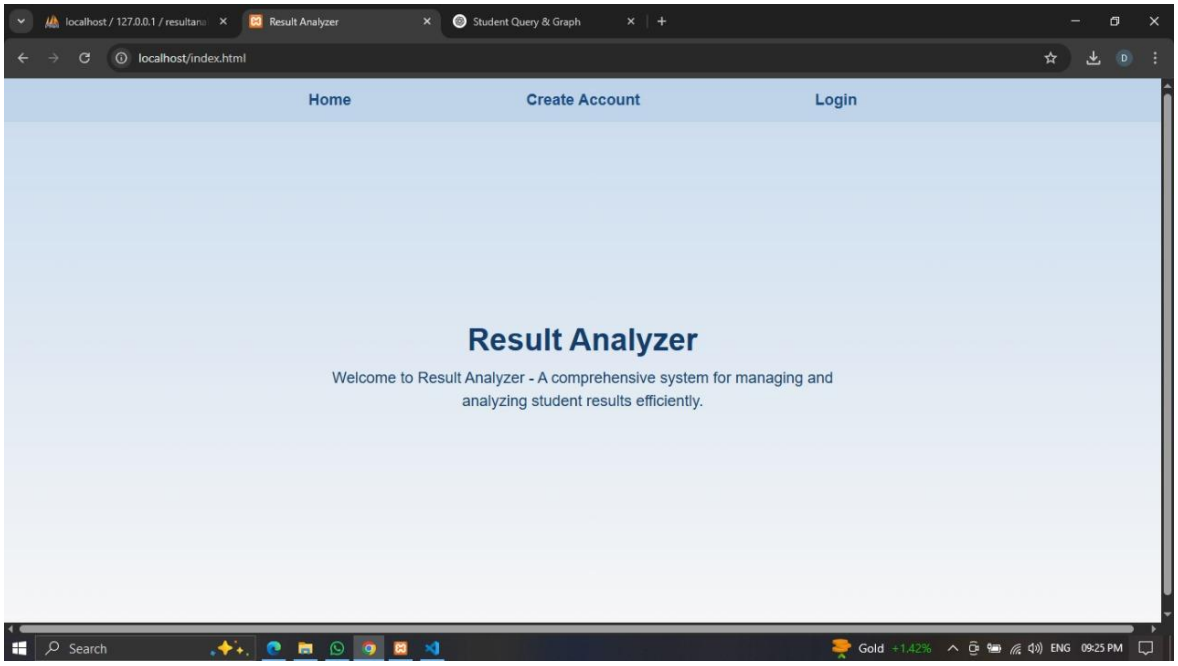


Figure 7.1.1: Index Page

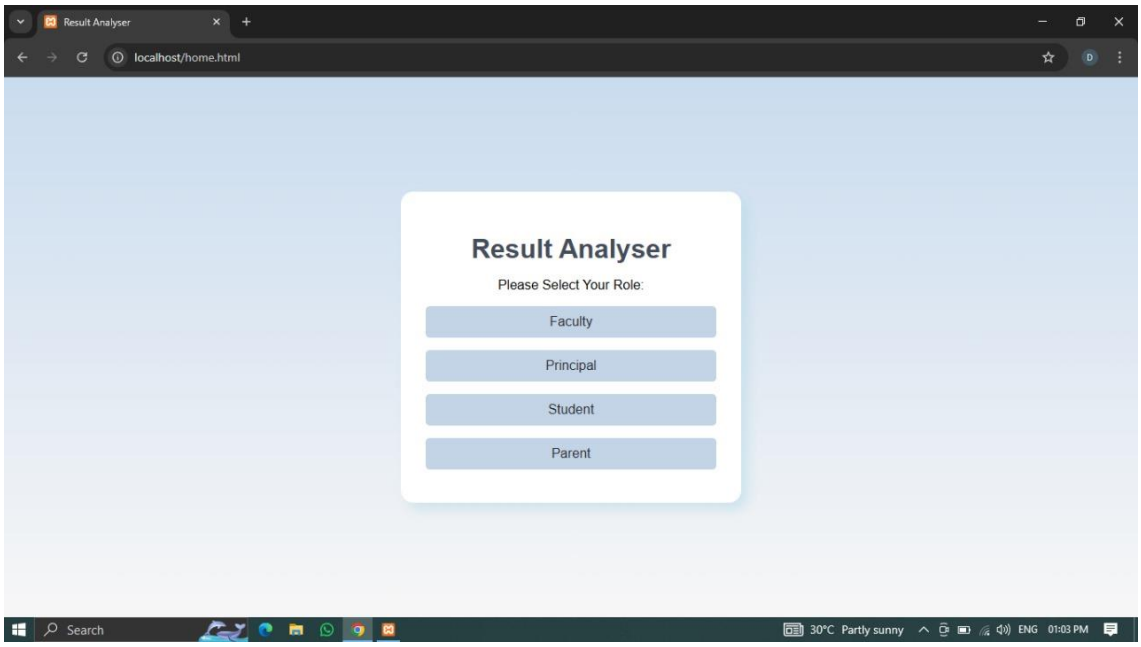


Figure 7.1.2: Home Page

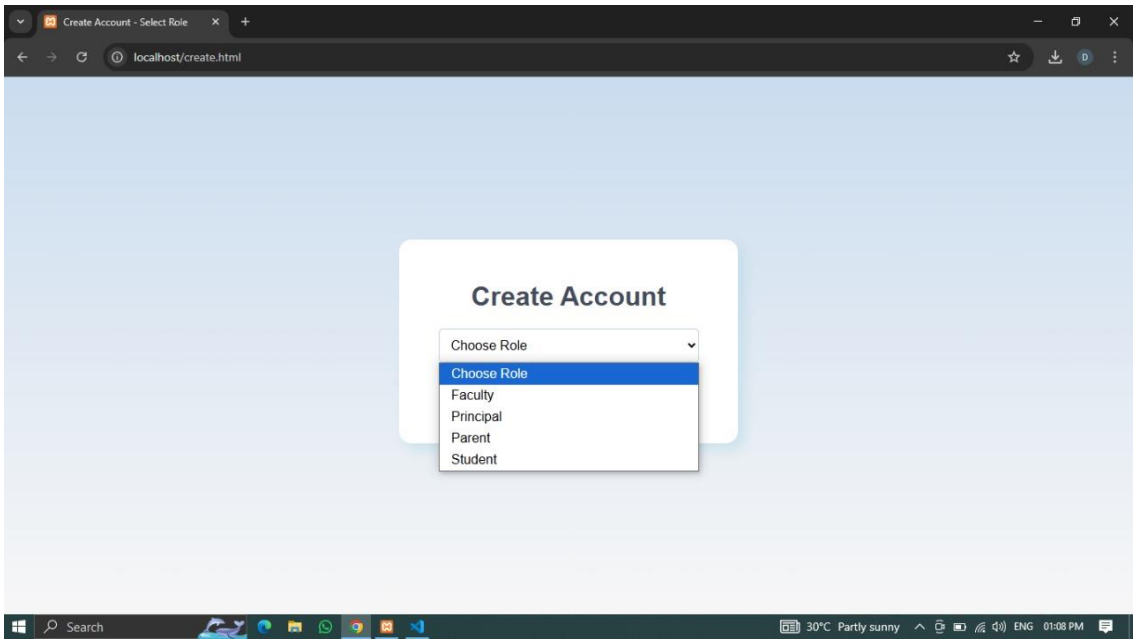


Figure 7.1.3: Account Creation Page

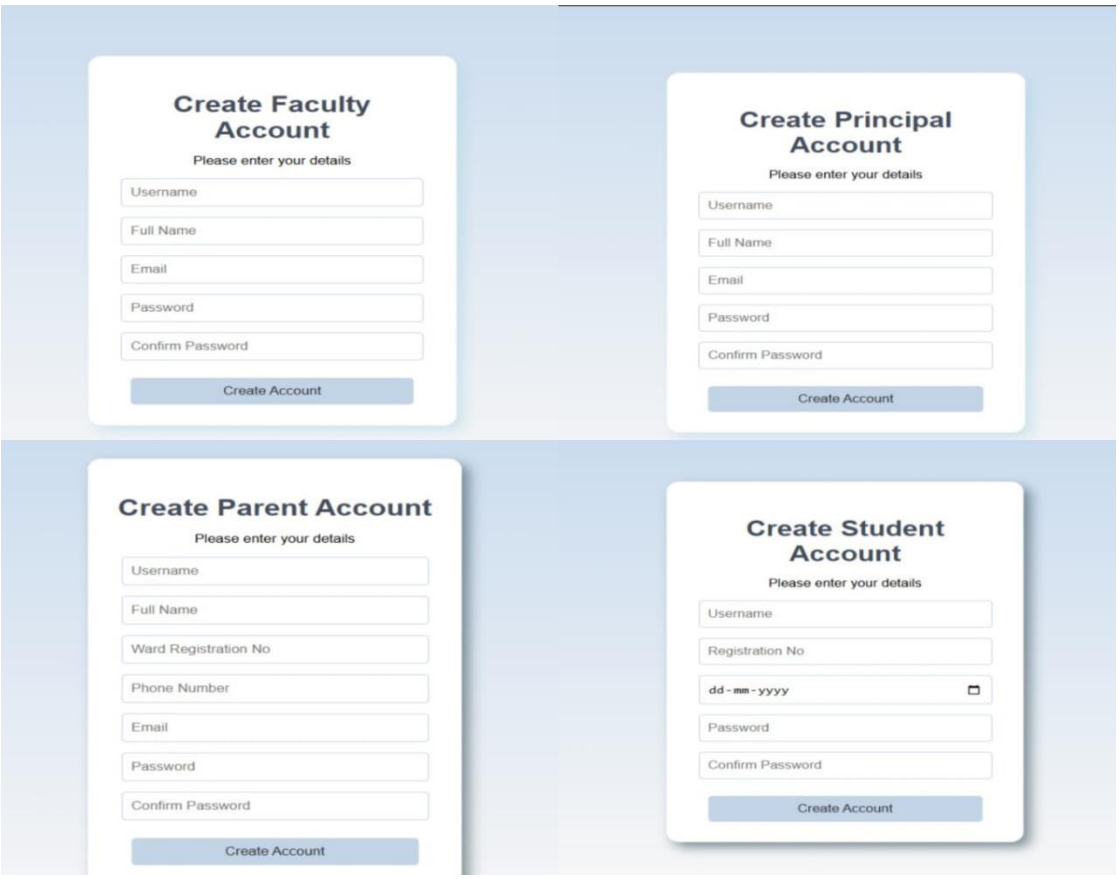


Figure 7.1.4: Role-Based Account Creation Page

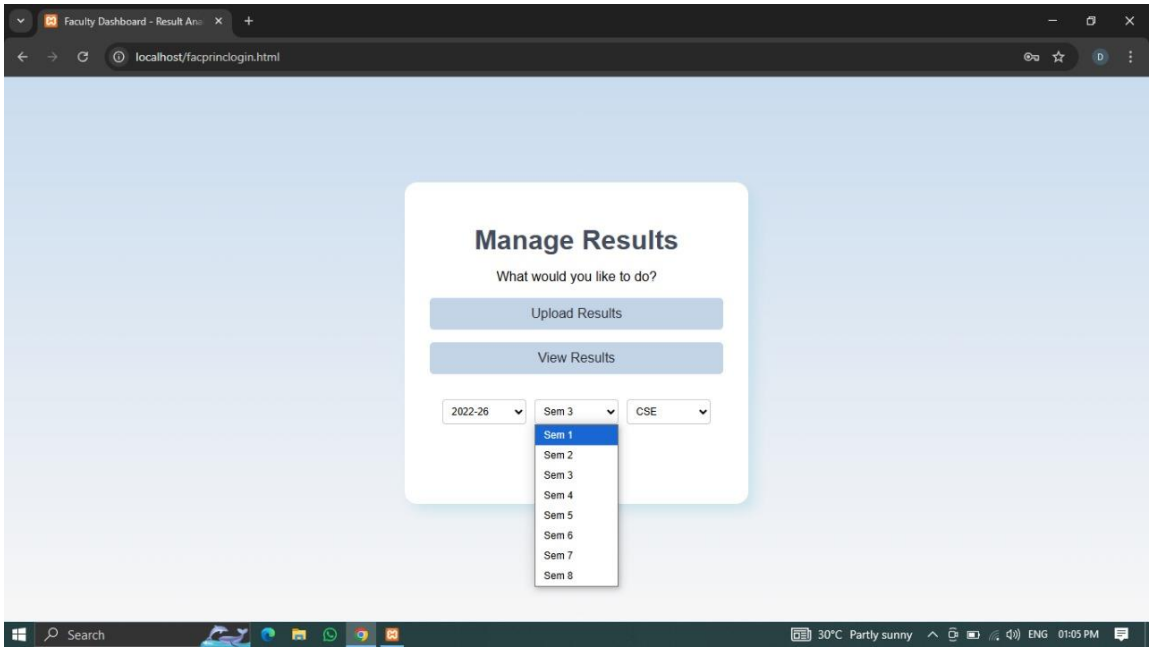


Figure 7.1.5: Result Management Dashboard

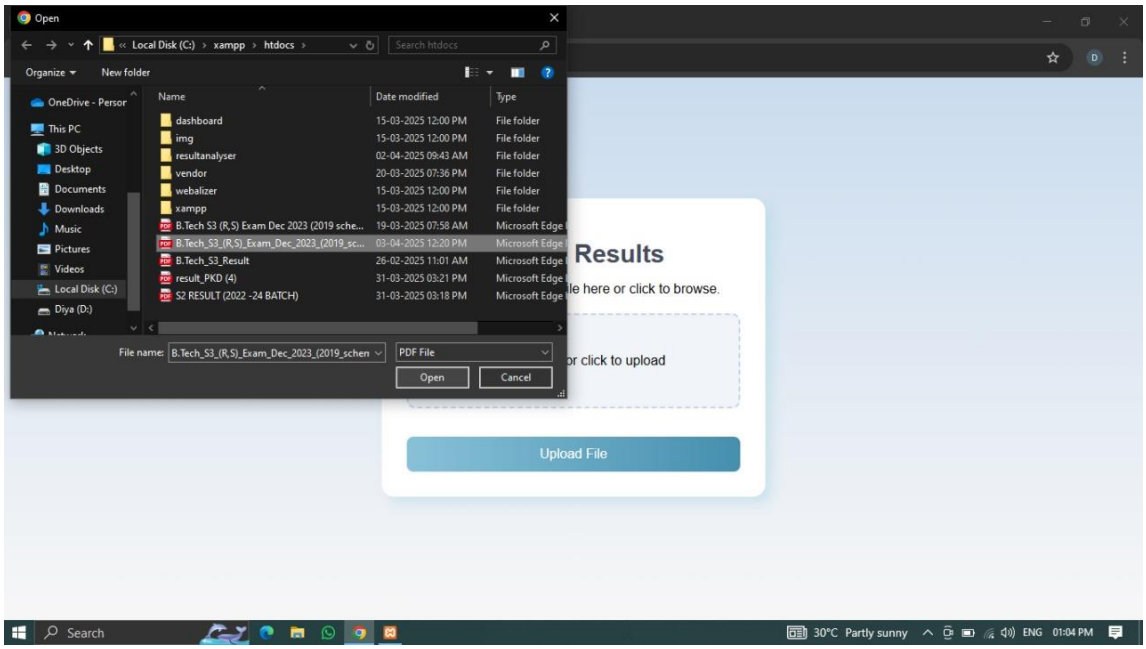
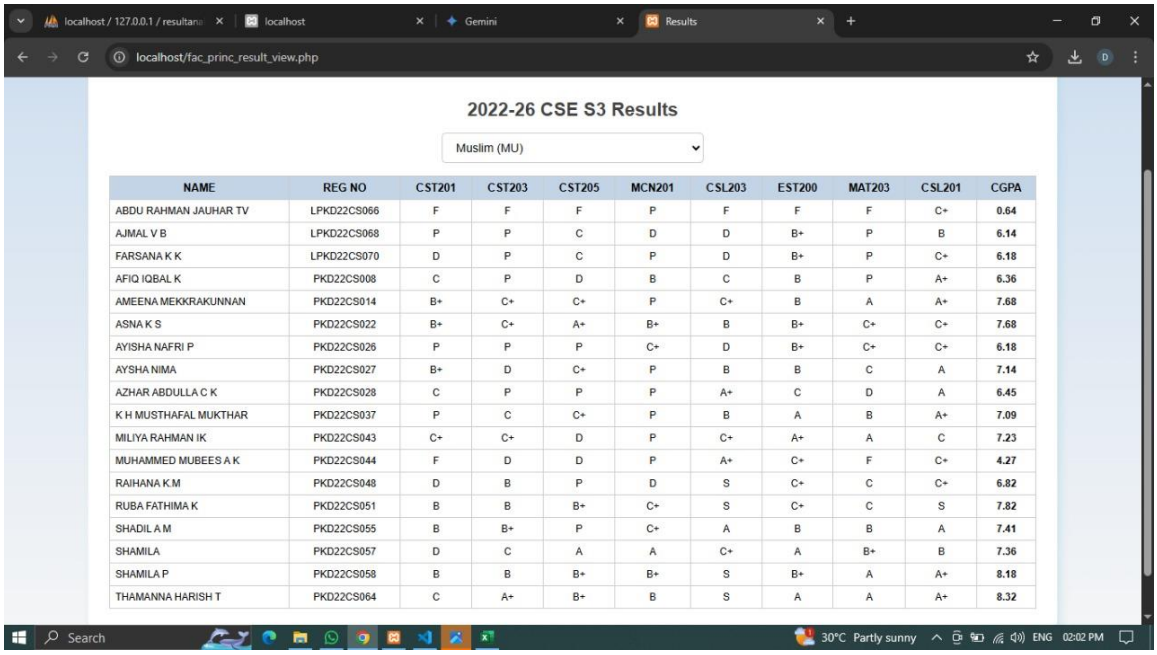


Figure 7.1.6: Uploading Results



NAME	REG NO	CST201	CST203	CST205	MCN201	CSL203	EST200	MAT203	CSL201	CGPA
ABDU RAHMAN JAUHAR TV	LPKD22CS066	F	F	F	P	F	F	F	C+	0.64
AJMAL V B	LPKD22CS068	P	P	C	D	D	B+	P	B	6.14
FARSANA K K	LPKD22CS070	D	P	C	P	D	B+	P	C+	6.18
AFIQ IQBAL K	PKD22CS008	C	P	D	B	C	B	P	A+	6.36
AMEENA MEKKRAKUNNAN	PKD22CS014	B+	C+	C+	P	C+	B	A	A+	7.68
ASNA K S	PKD22CS022	B+	C+	A+	B+	B	B+	C+	C+	7.68
AYISHA NAFRI P	PKD22CS026	P	P	P	C+	D	B+	C+	C+	6.18
AYSHA NIMA	PKD22CS027	B+	D	C+	P	B	B	C	A	7.14
AZHAR ABDULLA C K	PKD22CS028	C	P	P	P	A+	C	D	A	6.45
K H MUSTHAFAL MUKTHAR	PKD22CS037	P	C	C+	P	B	A	B	A+	7.09
MILIYA RAHMAN IK	PKD22CS043	C+	C+	D	P	C+	A+	A	C	7.23
MUHAMMED MUBEES A K	PKD22CS044	F	D	D	P	A+	C+	F	C+	4.27
RAIHANA K.M	PKD22CS048	D	B	P	D	S	C+	C	C+	6.82
RUBA FATHIMA K	PKD22CS051	B	B	B+	C+	S	C+	C	S	7.82
SHADIL A M	PKD22CS055	B	B+	P	C+	A	B	B	A	7.41
SHAMILA	PKD22CS057	D	C	A	A	C+	A	B+	B	7.36
SHAMILA P	PKD22CS058	B	B	B+	B+	S	B+	A	A+	8.18
THAMANNA HARISH T	PKD22CS064	C	A+	B+	B	S	A	A	A+	8.32

Figure 7.1.7: Results- Category Sorted Table View

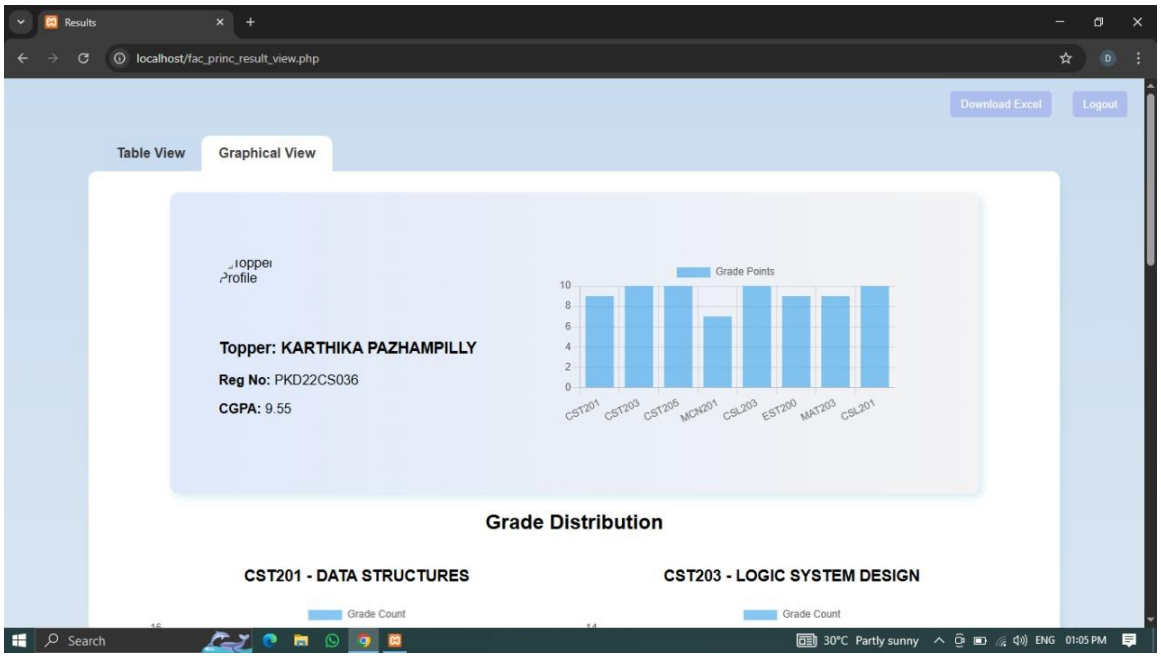


Figure 7.1.8: Results- Graphical View

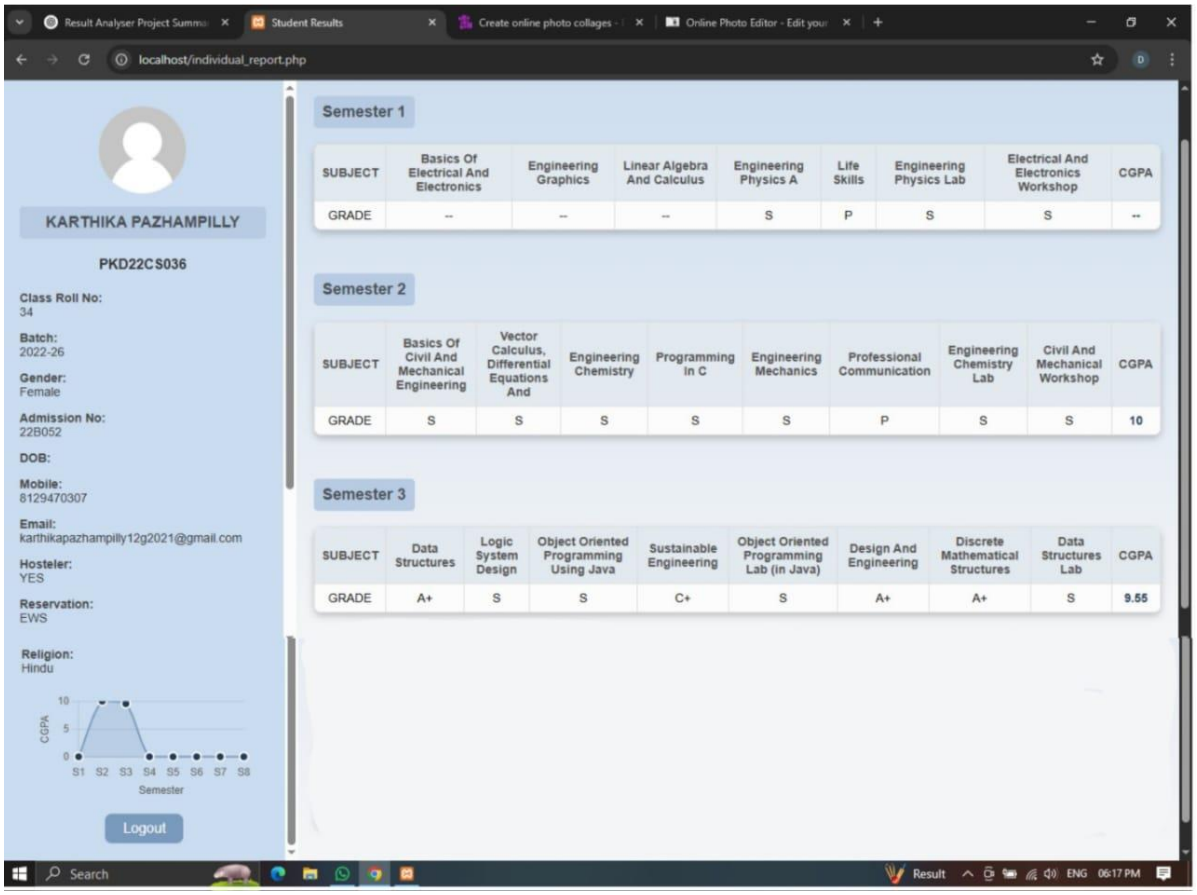


Figure 7.1.9: Individual Student Report
(Faculty/Principal)

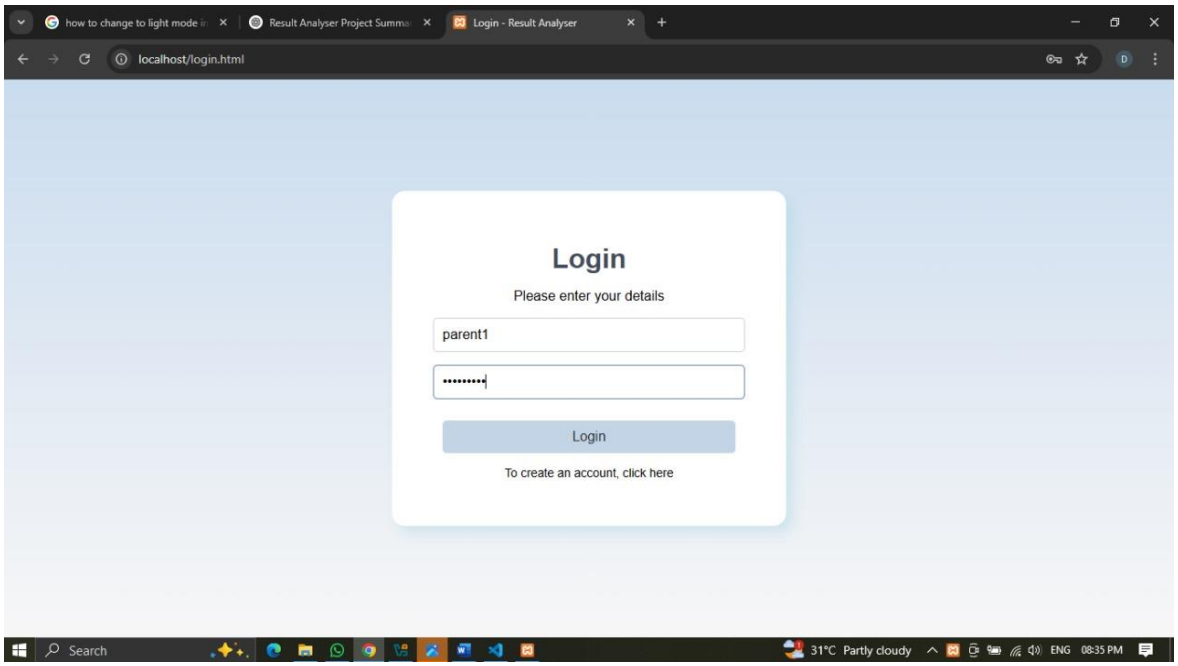


Figure 7.1.10: Parent Login

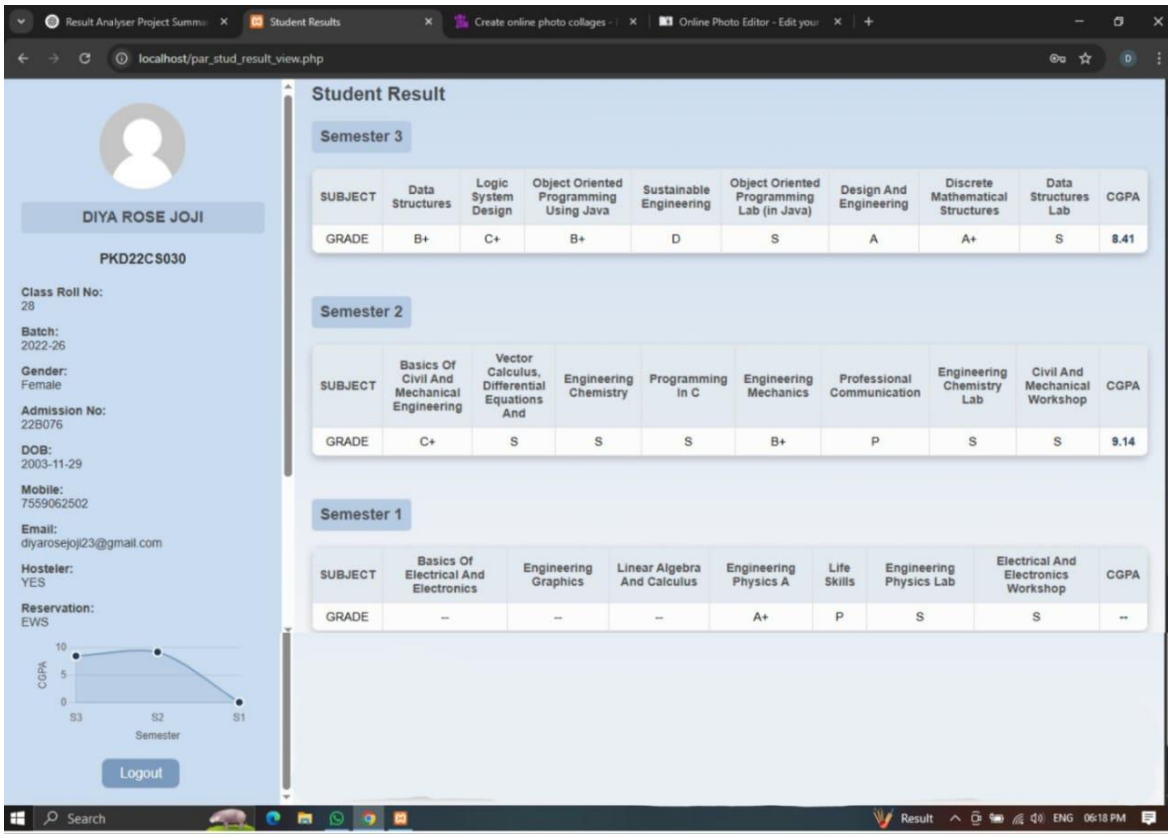
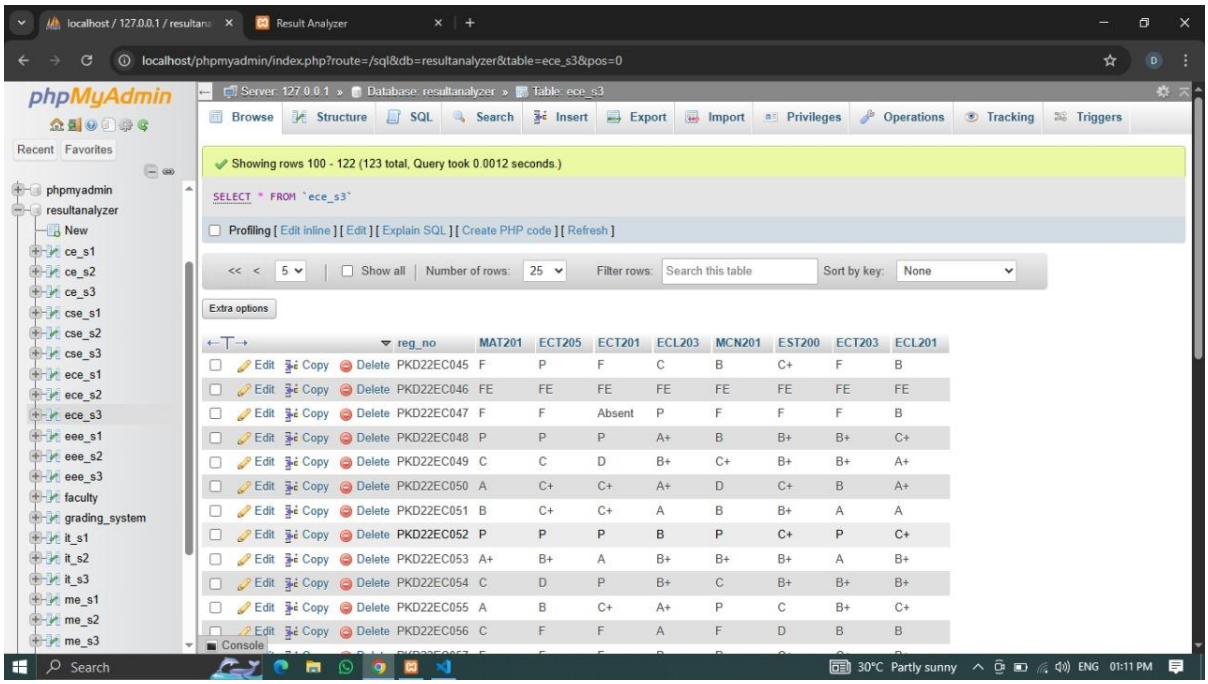


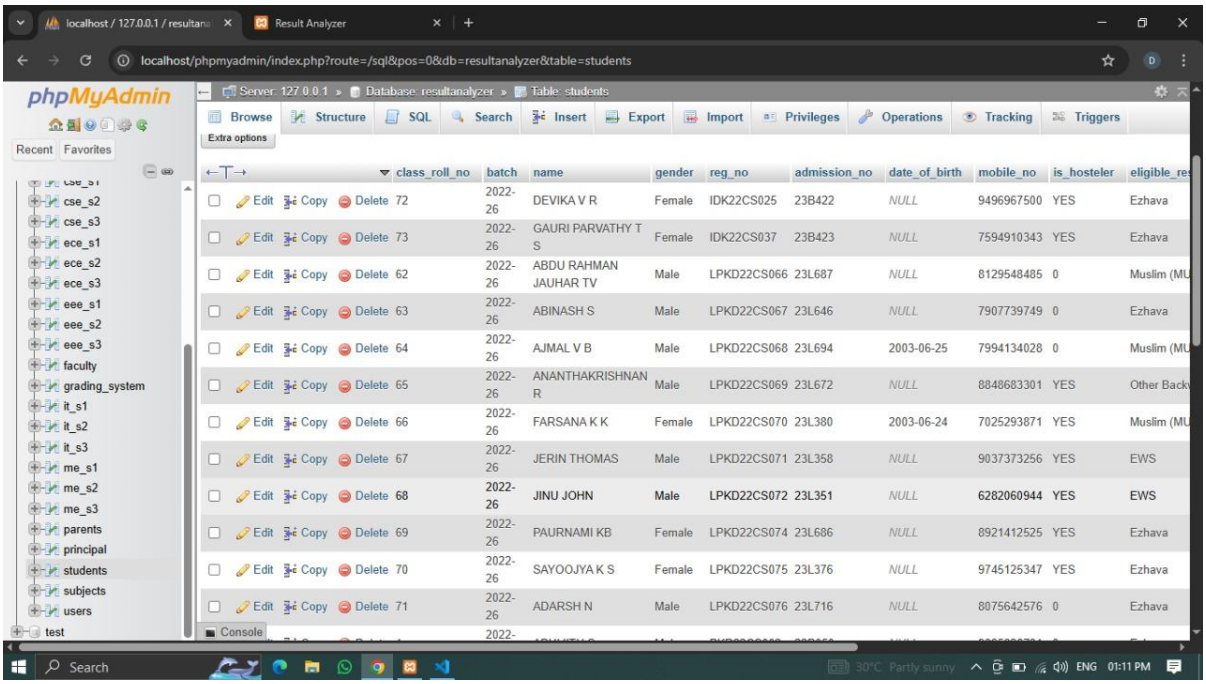
Figure 7.1.11: Individual Student Report
(Parent/Student)

DATABASES



reg_no	MAT201	ECT205	ECT201	ECL203	MCN201	EST200	ECT203	ECL201
PKD22EC045	F	P	F	C	B	C+	F	B
PKD22EC046	FE	FE	FE	FE	FE	FE	FE	FE
PKD22EC047	F	F	Absent	P	F	F	F	B
PKD22EC048	P	P	P	A+	B	B+	B+	C+
PKD22EC049	C	C	D	B+	C+	B+	B+	A+
PKD22EC050	A	C+	C+	A+	D	C+	B	A+
PKD22EC051	B	C+	C+	A	B	B+	A	A
PKD22EC052	P	P	P	B	P	C+	P	C+
PKD22EC053	A+	B+	A	B+	B+	B+	A	B+
PKD22EC054	C	D	P	B+	C	B+	B+	B+
PKD22EC055	A	B	C+	A+	P	C	B+	C+
PKD22EC056	C	F	F	A	F	D	B	B

7.1.12: Departmental Semester-wise Results table



class_roll_no	batch	name	gender	reg_no	admission_no	date_of_birth	mobile_no	is_hosteler	eligible_res
72	2022-26	DEVIKA V R	Female	IDK22CS025	23B422	NULL	9496967500	YES	Ezhava
73	2022-26	GAURI PARVATHY T S	Female	IDK22CS037	23B423	NULL	7594910343	YES	Ezhava
62	2022-26	ABDU RAHMAN JAUHAR TV	Male	LPKD22CS066	23L687	NULL	8129548485	0	Muslim (MU
63	2022-26	ABINASH S	Male	LPKD22CS067	23L646	NULL	7907739749	0	Ezhava
64	2022-26	AJMAL V B	Male	LPKD22CS068	23L694	2003-06-25	7994134028	0	Muslim (MU
65	2022-26	ANANTHAKRISHNAN R	Male	LPKD22CS069	23L672	NULL	8848683301	YES	Other Backw
66	2022-26	FARSANA K K	Female	LPKD22CS070	23L380	2003-06-24	7025293871	YES	Muslim (MU
67	2022-26	JERIN THOMAS	Male	LPKD22CS071	23L358	NULL	9037373256	YES	EWS
68	2022-26	JINU JOHN	Male	LPKD22CS072	23L351	NULL	6282060944	YES	EWS
69	2022-26	PAURNAMI KB	Female	LPKD22CS074	23L686	NULL	8921412525	YES	Ezhava
70	2022-26	SAYOOJYA K S	Female	LPKD22CS075	23L376	NULL	9745125347	YES	Ezhava
71	2022-26	ADARSH N	Male	LPKD22CS076	23L716	NULL	8075642576	0	Ezhava

7.1.13: Students table

7.2 DISCUSSION

The implementation of the **Student Result Analyzer** showcased the value of a structured and modular system in managing academic performance data efficiently. The integration of result uploading, filtering, and dynamic reporting modules demonstrated how well-organized architecture can streamline institutional result analysis without the need for complex AI models.

Some challenges encountered included:
• Handling inconsistent data formats during CSV uploads.
• Ensuring accuracy in data mapping across different departments and semesters.
• Managing system responsiveness when dealing with large datasets.

To address these issues, data validation checks were introduced during file uploads, the database schema was normalized for consistency, and query optimization techniques were applied to improve load times.

Future enhancements may include features like role-based access control, automated notifications for result publishing, and historical report generation for alumni tracking. These updates would further increase the platform's utility for administrators, faculty, and students alike.

CHAPTER 8
CONCLUSION

The *Student Result Analyzer* project marks a significant step toward modernizing academic performance evaluation in educational institutions. By integrating web technologies like HTML, CSS, JavaScript, PHP, and MySQL, the system offers a robust platform that enables seamless management and analysis of student results. This tool bridges the communication gap between teachers, parents, and students by providing a centralized, accessible, and interactive interface for performance tracking.

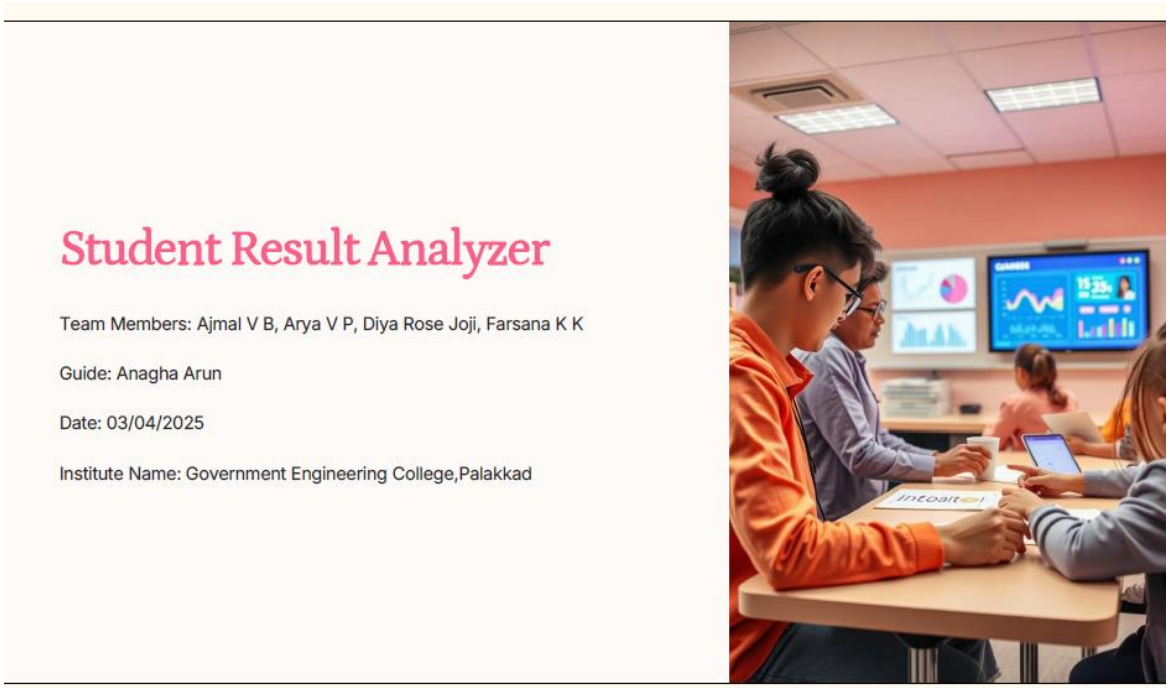
Through features such as secure authentication, result input and update modules, and graphical performance visualization using Chart.js, the system empowers educators to identify academic trends and offer timely interventions. At the same time, students and parents gain clarity on academic standing and growth, promoting transparency and motivation.

Challenges such as database integration, security concerns, and effective data visualization were successfully tackled, leading to a functional, scalable system. Furthermore, the modular design ensures that future enhancements—like AI-powered performance predictions, automated grading, and mobile accessibility—can be easily integrated.

In summary, this project not only reduces the workload of result analysis but also transforms raw data into meaningful insights, fostering a data-driven educational environment. With continued development, the *Student Result Analyzer* holds great potential to become an essential tool for academic institutions striving for excellence.

APPENDIX-A

A.1 PRESENTATION



Objectives: Streamlining Academic Management

1. Automate result management
2. Secure login system
3. Progress reports for all
4. Pdf to Excel conversion
5. Visualize performance

Key Features: Empowering Stakeholders

Teachers

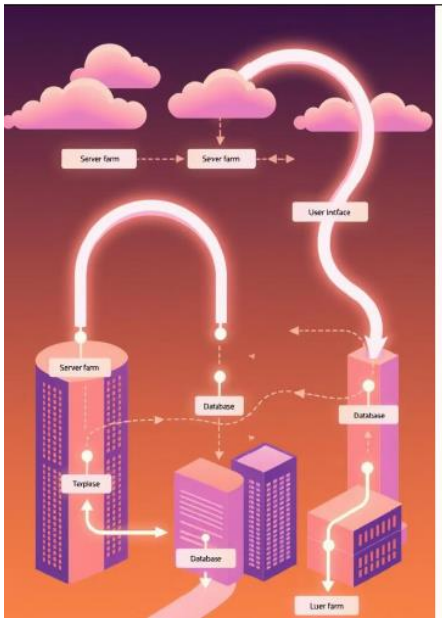
- Secure login
- Add/update results
- Analyze performance

Parents & Students

- View results
- Track progress
- Visual graphs

Technology Stack: Building the Future

1. **Frontend**
   HTML, CSS, JavaScript
2. **Backend**
   PHP
3. **Database**
   MySQL
4. **Tools**
   Chart.js, Bootstrap



System Architecture: Workflow Visualization

1. Login
2. Database
3. Teacher/Student Panel
4. Visualized Reports

Module Partition: Teamwork in Action

Authentication

Login page development

PHP & MySQL security

Result Management

Add/update student marks

Secure data storage

Visualization & Analysis

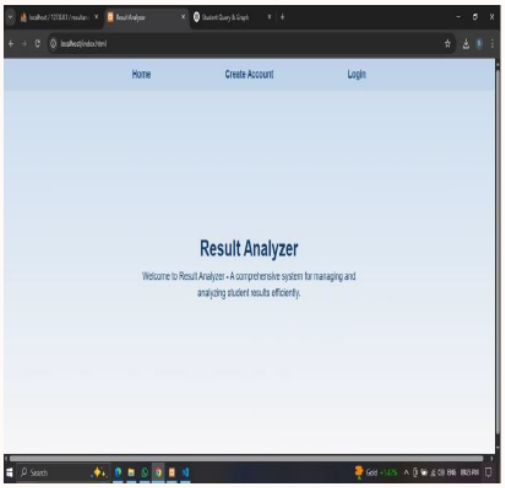
Chart.js for graphs

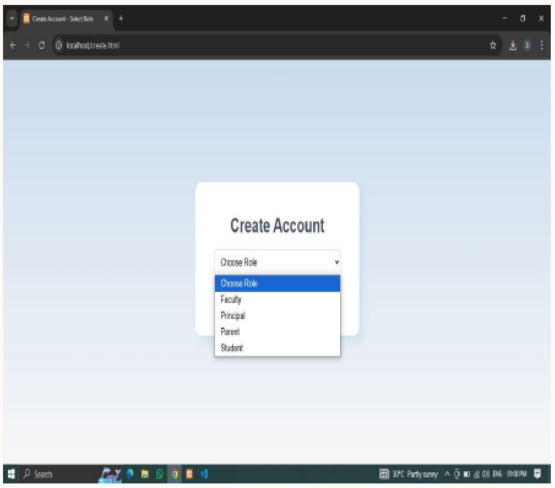
Performance trends insights

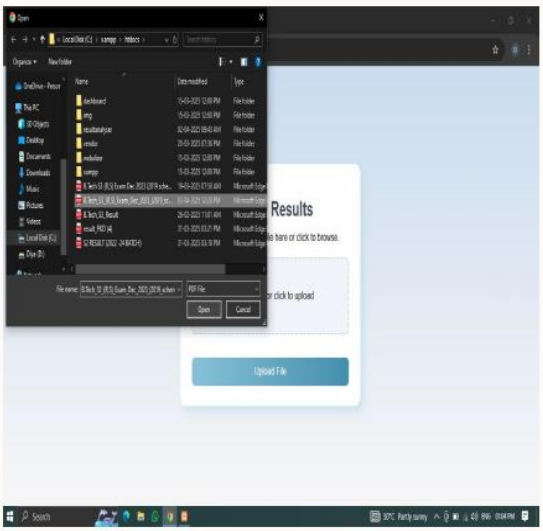
UI/UX & Testing

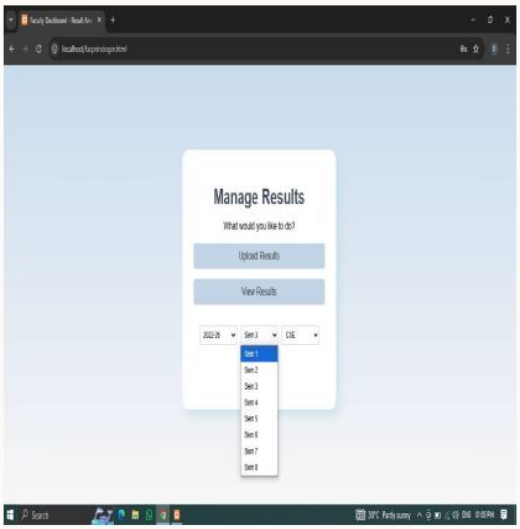
User-friendly interface

Smooth user experience






Results
Upload File


Manage Results
What would you like to do?
Upload Results
View Results

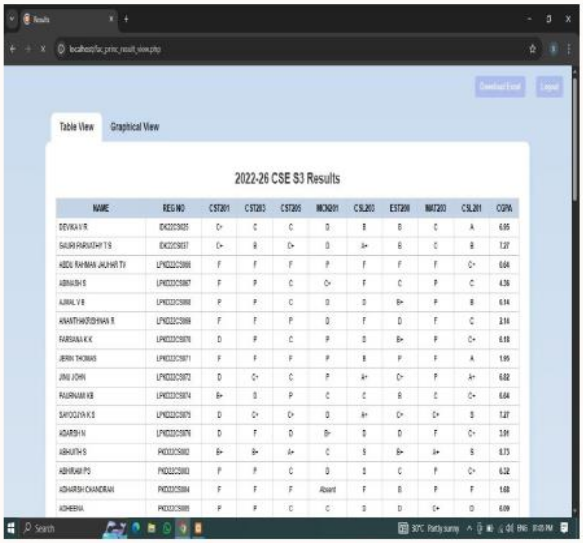


Table View
Graphical View
2022-26 CSE S3 Results

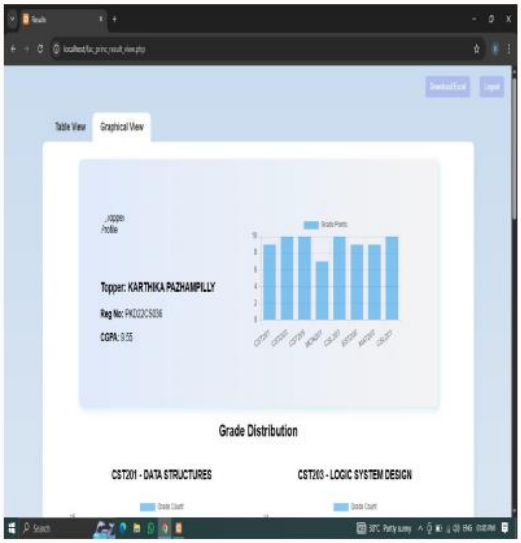
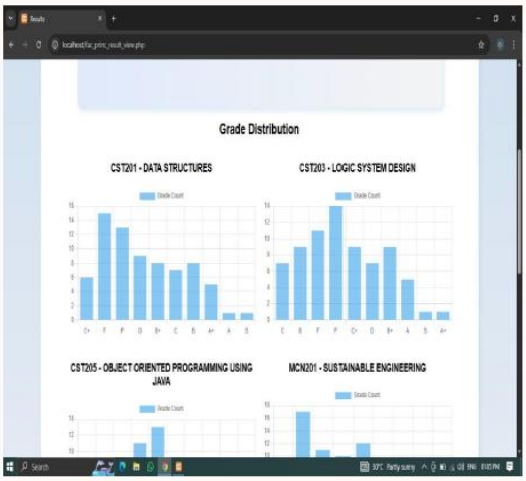
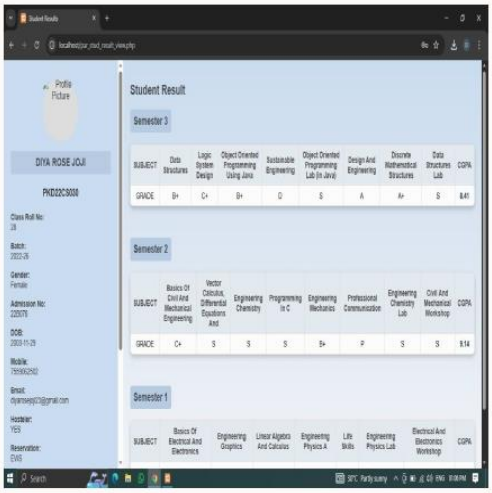
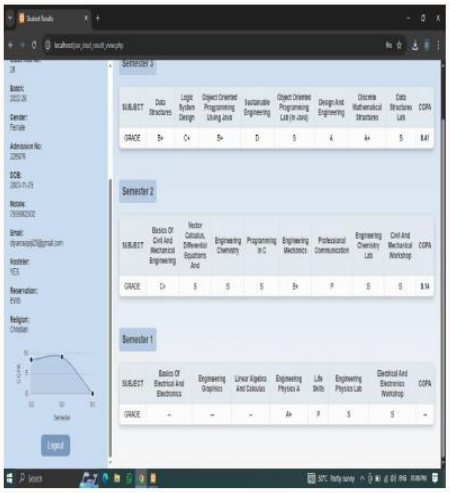


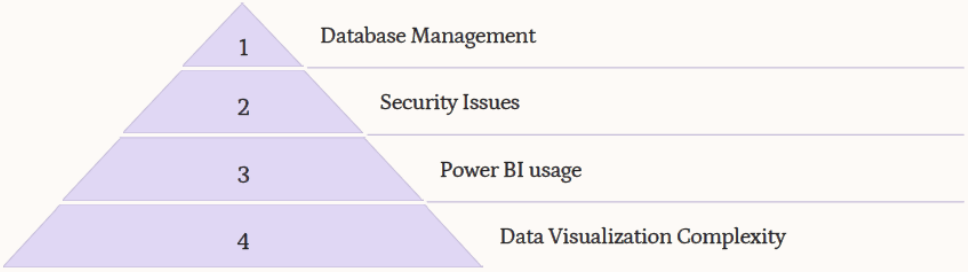
Table View
Graphical View
Topper: KARTHIKA PAZHAMPILLY
Grade Distribution
CST201 - DATA STRUCTURES
CST203 - LOGIC SYSTEM DESIGN


Grade Distribution
CST201 - DATA STRUCTURES
CST203 - LOGIC SYSTEM DESIGN
CST205 - OBJECT ORIENTED PROGRAMMING USING JAVA
MCN201 - SUSTAINABLE ENGINEERING

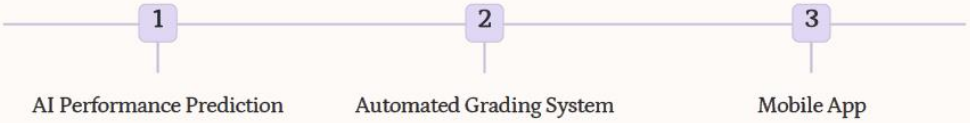

Student Result
Semester 3
Semester 2
Semester 1
DIYA ROSE JOJI


Semester 3
Semester 2
Semester 1
Logout

Challenges Faced: Overcoming Obstacles



Future Enhancements: Scaling New Heights



Conclusion

The **Result Analyzer** project effectively streamlines student performance tracking by providing a user-friendly platform for teachers, parents, and students. Teachers can easily **manage and analyze student results**, while parents and students can **track academic progress** through visual reports. The use of **HTML, CSS, PHP, and JavaScript** ensures a dynamic and interactive system. This project not only **reduces manual effort** but also enhances **data accuracy and accessibility**. Future enhancements like **AI-based performance prediction and mobile integration** can further improve the system's effectiveness.

Thank you

Thank you for your time and attention! We hope you found this presentation informative and insightful. We are open to any questions you may have.

APPENDIX-B

B.1 CODE SNIPPETS

1.Result Extraction:

```
<?php
session_start();
require 'vendor/autoload.php';
use PhpOffice\PhpSpreadsheet\Spreadsheet;
use PhpOffice\PhpSpreadsheet\Writer\Xlsx;

$servername = "localhost";
$username = "root";
$password = "";
$dbname = "resultanalyzer";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) die("❌ Database Connection Failed: "
. $conn->connect_error);

$batch = $_GET['batch'] ?? '';
$semester = $_GET['semester'] ?? '';
$department = $_GET['department'] ?? '';
if (empty($batch) || empty($semester) || empty($department))
die("❌ Missing required fields.");

$table_name = strtolower($department) . "_s" . $semester;
if (!preg_match('/^[a-z]+_s[0-9]+$/', $table_name)) die("❌
Invalid table name.");
if ($conn->query("SHOW TABLES LIKE '$table_name'")->num_rows ==
0) die("❌ Table $table_name does not exist.");


// Fetch grading system
$grade_map = [];
$res = $conn->query("SELECT grade, grade_point FROM
grading_system");
while ($row = $res->fetch_assoc()) $grade_map[$row['grade']] =
$row['grade_point'];


// Fetch subject credits once
$subject_credits = [];
$res = $conn->query("SELECT subject_code, credits FROM
subjects");
while ($row = $res->fetch_assoc())
$subject_credits[$row['subject_code']] = $row['credits'];


// Fetch results
$batch_prefix = substr($batch, 2, 2);
$sql = "SELECT s.name AS NAME, r.reg_no AS REG_NO, r.* FROM
$table_name r JOIN students s ON r.reg_no = s.reg_no WHERE
r.reg_no REGEXP '^[A-Za-z]+{$batch_prefix}[A-Za-z]+[0-9]+';";
$result = $conn->query($sql);
```

```
$rows = $result->fetch_all(MYSQLI_ASSOC);
if (empty($rows)) die("❌ No data found.");


// Compute CGPA
foreach ($rows as &$row) {
    $total_points = 0;
    $total_credits = 0;
    unset($row['REG_NO']);

    foreach ($row as $subject => $grade) {
        if    (!in_array($subject,    ['NAME',    'CGPA'])    &&
preg_match('/^[A-Za-z0-9]+$/', $grade)) {
            $gp = $grade_map[strtoupper($grade)] ?? 0;
            $credits = $subject_credits[$subject] ?? 3;
            $total_points += $gp * $credits;
            $total_credits += $credits;
        }
    }

    $row['CGPA'] = ($total_credits > 0) ? round($total_points /
$total_credits, 2) : "N/A";
}


// Prepare Excel
$spreadsheet = new Spreadsheet();
$sheet = $spreadsheet->getActiveSheet();
$sheet->setTitle(strtoupper($department)    .    "    S$semester
Results");
$headers = array_map('strtoupper', array_keys($rows[0]));
$sheet->fromArray([$headers], NULL, 'A1');
$sheet->getStyle("A1:"  .  chr(64  +  count($headers))  .  "1")-
>getFont()->setBold(true);
$sheet->freezePane('A2');


// Add data
$rowNum = 2;
foreach ($rows as $r) {
    $sheet->fromArray(array_values($r), NULL, "A$rowNum");
    $rowNum++;
}


// Output
header('Content-Type:           application/vnd.openxmlformats-
officedocument.spreadsheetml.sheet');
header('Content-Disposition: attachment; filename="Results_' .
$batch  .  '_'  .  strtoupper($department)  .  '_S'  .  $semester  .
'.xlsx"');
$writer = new Xlsx($spreadsheet);
$writer->save("php://output");
exit;
?>
```

2.Faculty/Principal Result View

```
// Database connection
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "resultanalyzer";
$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) die("❌ Database Connection Failed: "
. $conn->connect_error);


// Get form data
$batch = $_POST['batch'] ?? '';
$semester = $_POST['semester'] ?? '';
$department = $_POST['department'] ?? '';
$category_filter = $_POST['category_filter'] ?? '';
if (empty($batch) || empty($semester) || empty($department))
die("❌ Missing required fields.");


// Determine table name
$table_name = strtolower($department) . "_s" . $semester;
if ($conn->query("SHOW TABLES LIKE '$table_name'")->num_rows ==
0) die("❌ Table $table_name does not exist.");


// Fetch results
$batch_prefix = substr($batch, 2, 2);
$query = "SELECT s.name, s.reg_no,
s.eligible_reservation_category, r.*
          FROM $table_name r
          JOIN students s ON r.reg_no = s.reg_no
          WHERE r.reg_no REGEXP '^[A-Za-z]+{$batch_prefix}[A-
Za-z]+[0-9]+'";
if (!empty($category_filter)) {
    $query .= " AND s.eligible_reservation_category =
'$category_filter'";
}
$result = $conn->query($query);


// Fetch grading system and credits
$grades = [];
$res = $conn->query("SELECT grade, grade_point FROM
grading_system");
while ($row = $res->fetch_assoc()) $grades[$row['grade']] =
$row['grade_point'];

$credits = [];
$res = $conn->query("SELECT subject_code, credits FROM
subjects");
while ($row = $res->fetch_assoc())
$credits[$row['subject_code']] = $row['credits'];
```

```
// Function to calculate CGPA
function calculate_cgpa($row, $grades, $credits) {
    $total_credits = $total_score = 0;
    foreach ($row as $sub => $grade) {
        if (isset($grades[$grade], $credits[$sub])) {
            $total_score += $credits[$sub] * $grades[$grade];
            $total_credits += $credits[$sub];
        }
    }
    return  $total_credits  ?  "<b>"  .  round($total_score  /
$total_credits, 2) . "</b>" : '<b>--</b>';
}
```

3.Individual Result view:

```
$ward_reg_no = $_POST['reg_no'];

// Extract department code (e.g., PKD22CS030 → CS)
if (!preg_match('/[A-Z]{2}\d{2}([A-Z]{2})\d+/', $ward_reg_no,
$matches)) {
    die("❌ Invalid Registration Number Format!");
}

$dept_code = strtolower($matches[1]);
$dept_map = ["cs" => "cse", "ec" => "ece", "ee" => "eee", "it"
=> "it", "me" => "me", "ce" => "ce"];

if (!isset($dept_map[$dept_code])) {
    die("❌ Unknown department code: " . $dept_code);
}

$department = $dept_map[$dept_code];
$semesters = range(1, 8);
$all_results = [];
$subject_names = [];
$subject_credits = [];
$grade_points = [];


// Load grading system
$grade_result = $conn->query("SELECT grade, grade_point FROM
grading_system");
while ($grade = $grade_result->fetch_assoc()) {
    $grade_points[$grade['grade']] = $grade['grade_point'];
}

foreach ($semesters as $sem) {
    $table_name = "{$department}_s{$sem}";
    if ($conn->query("SHOW TABLES LIKE '$table_name';")-
>num_rows > 0) {
        $stmt = $conn->prepare("SELECT * FROM $table_name WHERE
reg_no = ?");
        $stmt->bind_param("s", $ward_reg_no);
        $stmt->execute();
        $result = $stmt->get_result();
```

```
        if ($row = $result->fetch_assoc()) {
            unset($row['reg_no']);
            foreach ($row as $sub_code => &$grade) {
                if (!in_array($grade, ["S", "A+", "A", "B+",
"B", "C+", "C", "D", "P", "F"])) {
                    $grade = "--";
                }
            }
            $all_results[$sem] = $row;


            // Subject details
            $subject_codes = array_keys($row);
            $placeholders = implode(',', array_fill(0,
count($subject_codes), '?'));
            $query = "SELECT subject_code, subject_name,
credits FROM subjects WHERE subject_code IN ($placeholders)";
            $stmt_sub = $conn->prepare($query);
            $stmt_sub->bind_param(str_repeat('s',
count($subject_codes)), ...$subject_codes);
            $stmt_sub->execute();
            $result_sub = $stmt_sub->get_result();
            while ($sub = $result_sub->fetch_assoc()) {
                $subject_names[$sub['subject_code']] =
$sub['subject_name'];
                $subject_credits[$sub['subject_code']] =
$sub['credits'];
            }


            // CGPA Calculation
            $total_points = 0;
            $total_credits = 0;
            foreach ($row as $sub_code => $grade) {
                if (isset($grade_points[$grade]) &&
isset($subject_credits[$sub_code])) {
                    $total_points += $grade_points[$grade] *
$subject_credits[$sub_code];
                    $total_credits +=
$subject_credits[$sub_code];
                }
            }
            $cgpa_data[$sem] = ($total_credits > 0) ?
round($total_points / $total_credits, 2) : 0;
        }
    }
}
```

REFERENCES

[1] W3Schools. (n.d.). *HTML, CSS, and JavaScript tutorials*. Retrieved from https://www.w3schools.com

[2] The PHP Group. (n.d.). *PHP manual*. Retrieved from https://www.php.net/manual/en/

[3] Oracle Corporation. (n.d.). *MySQL 8.0 Reference Manual*. Retrieved from https://dev.mysql.com/doc/

[4] Chart.js Contributors. (n.d.). *Chart.js: Simple yet flexible JavaScript charting*. Retrieved from https://www.chartjs.org/docs/latest/

[5] Singh, R., & Chauhan, M. (2020). Automated student result management system using PHP and MySQL. *International Journal of Computer Applications*, *176*(31), 20–24. https://doi.org/10.5120/ijca2020919942

[6] Sharma, P., & Patel, V. (2019). Data visualization for academic performance monitoring. In *Proceedings of the International Conference on Advances in Computing and Data Sciences (ICACDS)* (pp. 118–127). Springer. https://doi.org/10.1007/978-981-13-9942-8_11

[7] Bootstrap Team. (n.d.). *Bootstrap: Build fast, responsive sites*. Retrieved from https://getbootstrap.com

[8] Gamma Technologies. (2025). *Gamma: Visual storytelling platform for presentations*. Retrieved from https://gamma.app